

BTEC Level 3 Applied Science • Unit 1 Physics • Answer sheet

Wave Properties — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Transverse: oscillation perpendicular to energy transfer. Longitudinal: parallel.
2. Light / water / string; sound / P-waves / slinky compressions.
3. Resultant displacement is the sum of individual displacements.

B

4. Compression: particles closer / higher pressure. Rarefaction: more spaced / lower pressure.
5. They reach maxima and minima together / separated by $n\lambda$.
6. Light is transverse so oscillations have a direction that can be filtered; sound is longitudinal.

C

7. Ultrasound longitudinal; microwaves transverse; stadium sound longitudinal.
8. Yes — separation is 3λ , a whole number of wavelengths.
9. Destructive superposition / cancellation (resultant zero if amplitudes are equal).

D

10. See definitions plus one valid example of each.
11. Resultant displacement = sum of individual displacements.
12. Polarisation needs a transverse direction; sound is longitudinal.

E

13. The sine curve is a graph of pressure or displacement against distance or time. The air particles still oscillate parallel to the direction of travel.